MAT 230 EXAM TWO

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Directions: Type your solutions into this document and be sure to show all steps for arriving at your solution. Just giving a final number may not receive full credit.

Problem 1

This question has 2 parts.

Part 1: Suppose that F and X are events from a common sample space with $P(F) \neq 0$ and $P(X) \neq 0$.

(a) Prove that $P(X) = P(X|F)P(F) + P(X|\bar{F})P(\bar{F})$. Hint: Explain why $P(X|F)P(F) = P(X \cap F)$ is another way of writing the definition of conditional probability, and then use that with the logic from the proof of Theorem 4.1.1.

By law of total probability:

$$P(X) = P(X \cap F) + P(X \cap \bar{F})$$

$$P(X) = P(X|F)P(F) + P(X|\bar{F})P(\bar{F})$$

The first step uses the law of total probability. The second step applies the definition of conditional probability.

(b) Explain why $P(F|X) = P(X|F)P(F)/P(X)$ is another way of stating Theorem 4.2.1 Bayes Theorem.

$$P(F|X) = \frac{P(F \cap X)}{P(X)}$$

$$P(F|X) = \frac{P(X|F)P(F)}{P(X)}$$

This follows directly from the definition of conditional probability. Substituting this into the formula gives us Bayes' Theorem.

Part 2: A website reports that 70% of its users are from outside a certain country. Out of their users from outside the country, 60% of them log on every day. Out of their users from inside the country, 80% of them log on every day.

(a) What percent of all users log on every day? Hint: Use the equation from Part 1 (a).

Let F = user is from outside the country.
Let X = user logs on every day
Given: $P(F) = 0.7$, $P(X|F) = 0.6$, $P(X|\bar{F}) = 0.8$

$$P(X) = P(X|F)P(F) + P(X|\bar{F})P(\bar{F})$$

$$P(X) = 0.6 \times 0.7 + 0.8 \times 0.3$$

$$P(X) = 0.42 + 0.24 = 0.66$$

(b) Using Bayes Theorem, out of users who log on every day, what is the probability that they are from inside the country?

$$P(\bar{F}|X) = \frac{P(X|\bar{F})P(\bar{F})}{P(X)}$$

$$P(\bar{F}|X) = \frac{0.8 \times 0.3}{0.66} = \frac{0.24}{0.66} = \frac{24}{66} = \frac{4}{11} = 0.364$$

Approximately 36.4 percent

Problem 2

This question has 2 parts.

Part 1: The drawing below shows a Hasse diagram for a partial order on the set:
$\{A,\ B,\ C,\ D,\ E,\ F,\ G,\ H,\ I,\ J\}$

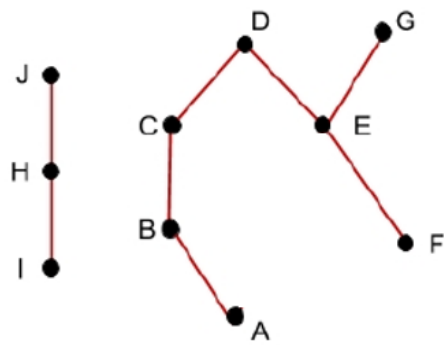


Figure 1: *A Hasse diagram shows 10 vertices and 8 edges. The vertices, represented by dots, are as follows: vertex J is upward of vertex H; vertex H is upward of vertex I; vertex B is inclined upward to the left of vertex A; vertex C is upward of vertex B; vertex D is inclined upward to the right of vertex C; vertex E is inclined upward to the left of vertex F; vertex G is inclined upward to the right of vertex E. The edges, represented by line segments between the vertices are as follows: 3 vertical edges connect the following vertices: B and C, H and I, and H and J; 5 inclined edges connect the following vertices: A and B, C and D, D and E, E and F, and E and G.*

Determine the properties of the Hasse diagram based on the following questions:

(a) What are the minimal elements of the partial order?

The minimal elements are A, I
Minimal elements have no elements below them in the Hasse diagram

(b) What are the maximal elements of the partial order?

The maximal elements are D, G, J, F
Maximal elements have no elements above them in the Hasse diagram.

(c) Which of the following pairs are comparable?

$$(A,\ D),\ (J,\ F),\ (B,\ E),\ (G,\ F),\ (D,\ B),\ (C,\ F),\ (H,\ I), (C,\ E)$$

The comparable pairs are (A, D), (B, E), (H, I), (C, E)
Pairs are comparable if there exists a path in the Hasse diagram from one element to the other.
(A,D): Path A-B-C-D
(B,E): Path B-C-D-E
(H,I): Direct edge from H to I
(C,E): Path C-D-E

Part 2: Consider the partial order with domain $\{3, 5, 6, 7, 10, 14, 20, 30, 60, 70\}$ and with $x \leq y$ if x evenly divides y. Select the correct Hasse diagram for the partial order.

(a)

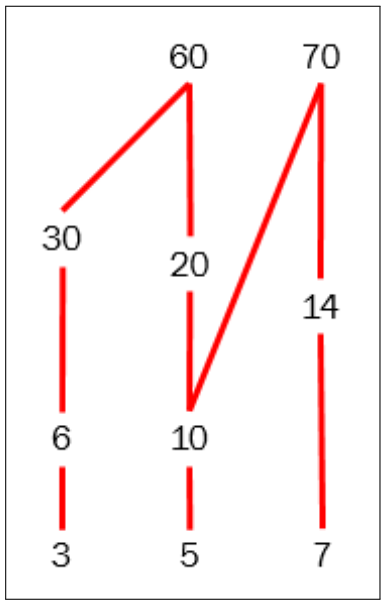


Figure 2: *A Hasse diagram shows a set of elements 3; 5; 6; 7; 10; 14; 20; 30; 60, 70. There are lines connecting 3 and 6, 6 and 30, 30 and 60, 5 and 10, 10 and 20, 20 and 60, 10 and 70, 7 and 14, 14 and 70.*

The correct Hasse diagram for the divisibility relation is option (b).

In option (b), the diagram correctly shows the divisibility relation. For example:

-3 divides 6, 6 divides 30, 30 divides 60

-5 divides 10, 10 divides 30, 10 divides 20, 20 divides 60

-7 divides 14, 14 divides 70

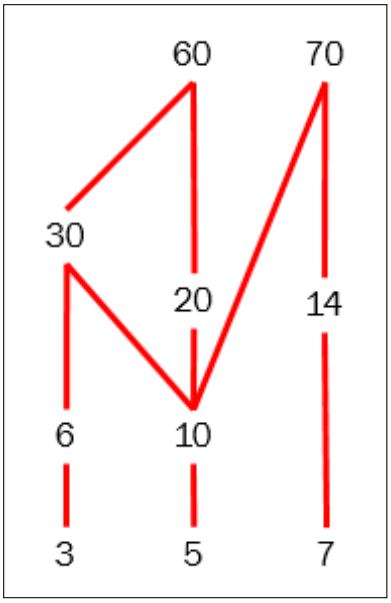
(b) 

Figure 3: *A Hasse diagram shows a set of elements 3; 5; 6; 7; 10; 14; 20; 30; 60, 70. There are lines connecting 3 and 6, 6 and 30, 30 and 60, 5 and 10, 10 and 30, 10 and 20, 20 and 60, 10 and 70, 7 and 14, 14 and 70.*

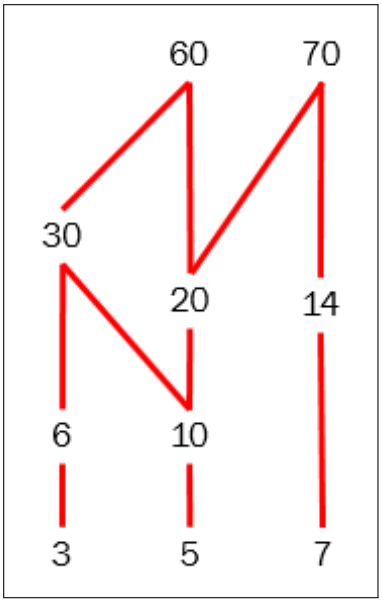
(c) 

Figure 4: *A Hasse diagram shows a set of elements 3; 5; 6; 7; 10; 14; 20; 30; 60, 70. There are lines connecting 3 and 6, 6 and 30, 30 and 60, 5 and 10, 10 and 30, 10 and 20, 20 and 60, 20 and 70, 7 and 14, 14 and 70.*

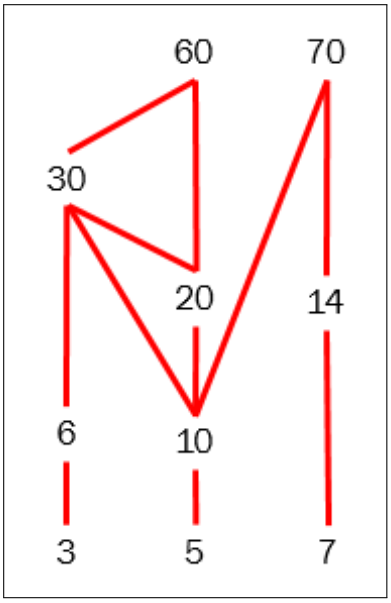
(d) 

Figure 5: *A Hasse diagram shows a set of elements 3; 5; 6; 7; 10; 14; 20; 30; 60, 70. There are lines connecting 3 and 6, 6 and 30, 30 and 60, 5 and 10, 10 and 30, 10 and 20, 20 and 30, 20 and 60, 10 and 70, 7 and 14, 14 and 70.*

Problem 3

A car dealership sells cars that were made in 2015 through 2020. Let the cars for sale be the domain of a relation R where two cars are related if they were made in the same year.

(a) Prove that this relation is an equivalence relation.

To prove R is an equivalence relation, I must show it's reflexive, symmetric, and transitive.
1. Reflexive: For any car c, c was made in the same year as itself, so (c,c)E R.

2. Symmetric: If car(c1) was made in the same year as car(c2), then car(c2) was made in the same year as car(c1). Thus, if(c1,c2)E R, then(c2,c1)E R.

3. Transitive: If car c1 was made in the same year as car c2, and car c2 was made in the same year as car c3, then c1 and c3 were made in the same year.

Therefore, R is an equivalence relation.

(b) Describe the partition defined by the equivalence classes.

The partition consists of 6 equivalence classes:

$$[2015] = \{\text{all cars made in 2015}\}$$
$$[2016] = \{\text{all cars made in 2016}\}$$
$$[2017] = \{\text{all cars made in 2017}\}$$
$$[2018] = \{\text{all cars made in 2018}\}$$
$$[2019] = \{\text{all cars made in 2019}\}$$
$$[2020] = \{\text{all cars made in 2020}\}$$

Problem 4

Analyze each graph below to determine whether it has an Euler circuit and/or an Euler trail.

- If it has an Euler circuit, specify the nodes for one.
- If it does not have an Euler circuit, justify why it does not.
- If it has an Euler trail, specify the nodes for one.
- If it does not have an Euler trail, justify why it does not.

(a)

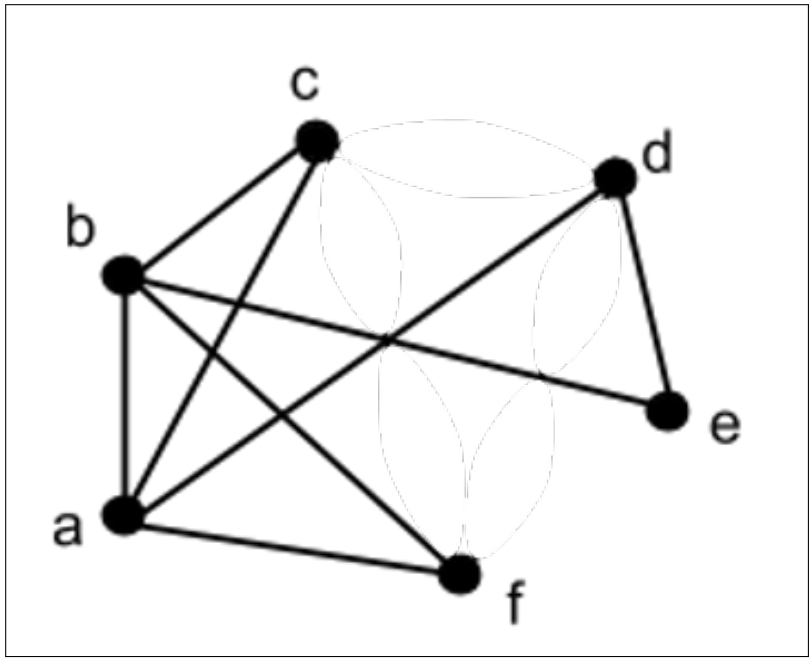


Figure 6: *An undirected graph has 6 vertices, a through f. There are 8-line segments that are between the following vertices: a and b, a and c, a and d, a and f, b and c, b and e, b and f, d and e.*

Euler circuit: No

Justification: Vertices c, d, and e have odd degree (3,3, and 3 respectively). A graph has an Euler circuit if and only if all vertices have an even degree.

Euler trail: No

Justification: A graph has an Euler trail if and only if it has exactly 2 vertices with odd degree, but this graph has 3 vertices with odd degree.

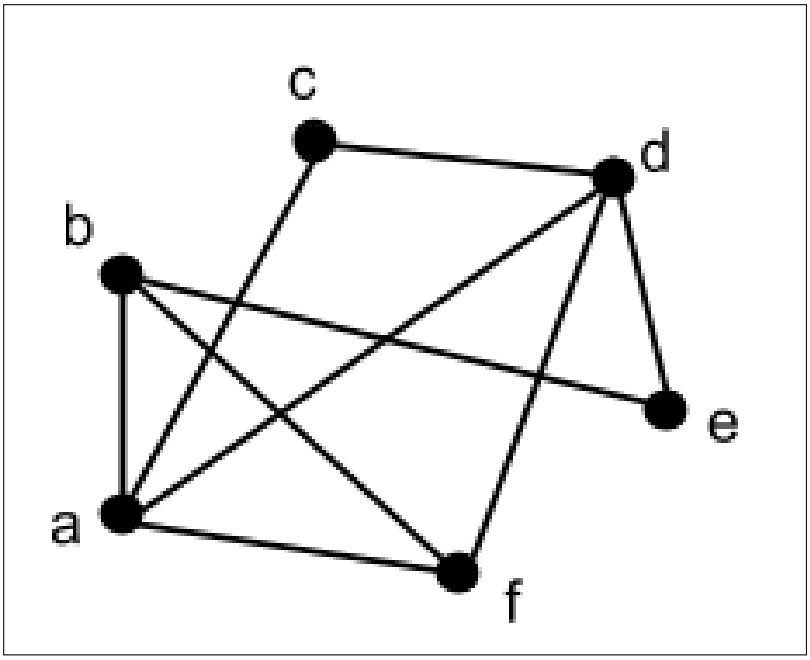
(b) 

Figure 7: *An undirected graph has 6 vertices, a through f. There are 9-line segments that are between the following vertices: a and b, a and c, a and d, a and f, b and e, b and f, c and d, d and e, d and f.*

Euler circuit: No

Justification: Vertices a, b, c, and d have odd degree. A graph has an Euler circuit if and only if all vertices have even degree.

Euler trail: No

Justification: A graph has an Euler trail if and only if it has exactly 2 vertices with odd degree, but this graph has 4 vertices with odd degree.

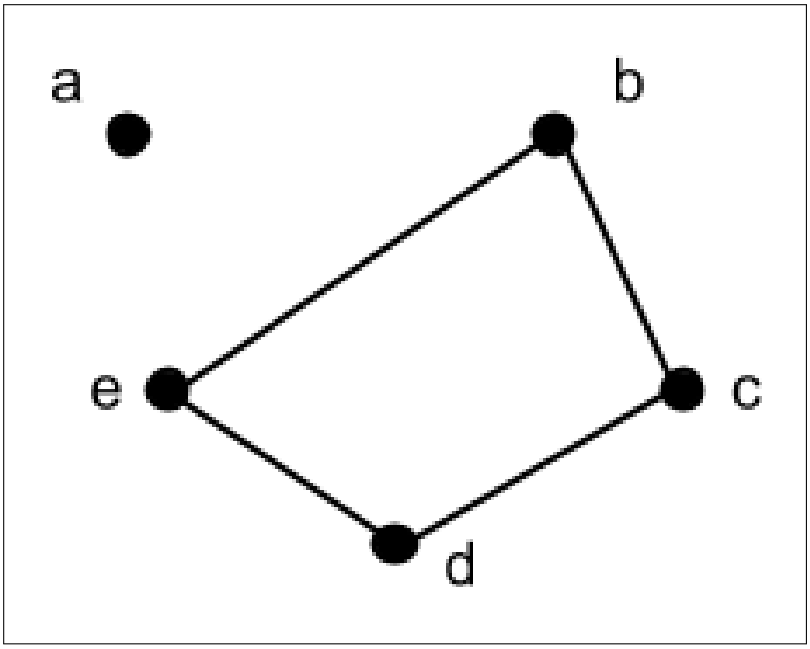
(c) 

Figure 8: *An undirected graph has 5 vertices, a through e. There are 4-line segments that are between the following vertices: b and c, b and e, c and d, d and e.*

Euler circuit: No

Justification: Vertices b, c, d, and e have odd degree. A graph has an Euler circuit if and only if all vertices have even degree.

Euler trail: No

Justification: A graph has an Euler trail if and only if it has exactly 2 vertices with odd degree, but this graph has 4 vertices with odd degree.

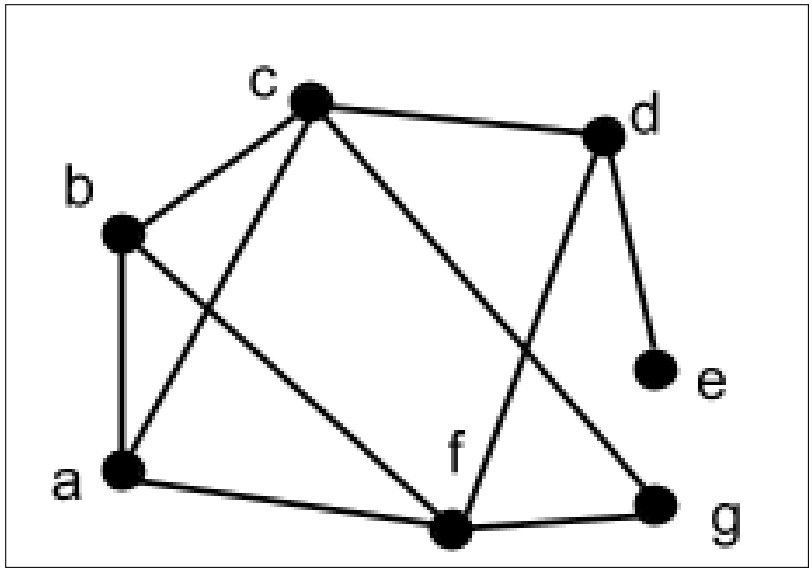
(d) 

Figure 9: *An undirected graph has 7 vertices, a through g. There are 10-line segments that are between the following vertices: a and b, a and c, a and f, b and c, b and f, c and d, c and g, d and e, d and f, f and g.*

Euler circuit: Yes

Justification: All vertices have even degree (4,4,4,4,2,4,2). One possible Euler circuit: $a - b - c - d - e - d - f - g - c - g - f - a$

Euler trail: Yes

Since the graph has an Euler circuit, it also has an Euler trail.

Problem 5

Use Prim's algorithm to compute the minimum spanning tree for the weighted graph. Start the algorithm at vertex A. Explain and justify each step as you add an edge to the tree.

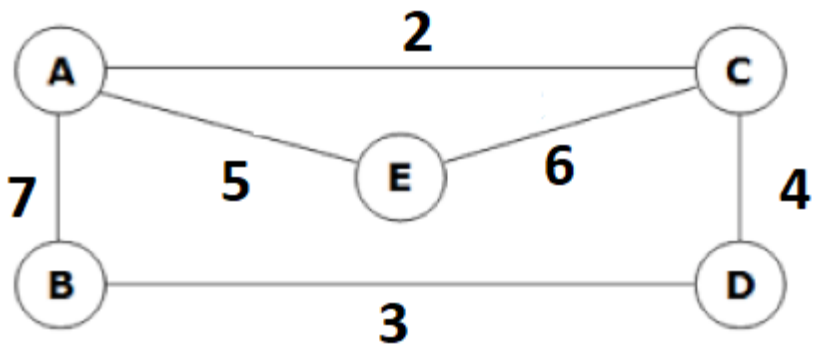


Figure 10: *A weighted graph shows 5 vertices, represented by circles, and 6 edges, represented by line segments. Vertices A, B, C, and D are placed at the corners of a rectangle, whereas vertex E is at the center of the rectangle. The edges, A B, B D, A C, C D, A E, and E C, have the weights, 7, 3, 2, 4, 5, and 6, respectively.*

Prim's Algorithm for Minimum Spanning Tree:

Step 1: Start with vertex A. Consider all edges from A: (A,B) with weight 7, (A,C) with weight 2, and (A,E) with weight 5. Select(A,C) as it has the minimum weight of 2.

Tree: A-C

Step 2: Vertices in tree: A,C. Consider all edges from A and C to vertices not in the tree (B,D,E): (A,B) with weight 7, (A,E) with weight 5, (C,D) with weight 4.

Tree: A-C-D

Step 3: Vertices in tree: A,C,D. Consider all edges from A, C, and D to vertices not in the tree(B,E): (A,B) with weight 7, (A,E) with weight 5, (C,E) with weight 6, (B,D) with weight 3. Select (B,D) with weight 3.

Tree: A-C-D-B

Step 4: Vertices in tree: A,C,D,B. Consider all edges from A,C,D, and B to vertices not in the tree(E): (A,E) with weight 5, (C,E) with weight 6. Select (A,E) with weight 5.

Tree: A-C-D-B, A-E

The minimum spanning tree has edges:

(A,C),(C,D),(D,B),(A,E) with a total weight of 2+4+3+5=14.

Problem 6

A lake initially contains 1000 fish. Suppose that in the absence of predators or other causes of removal, the fish population increases by 10% each month. However, factoring in all causes, 80 fish are lost each month.

Give a recurrence relation for the population of fish after n months. How many fish are there after 5 months? If your fish model predicts a non-integer number of fish, round down to the next lower integer.

Let a_n be the fish population after n months.

$$a_0 = 1000 \text{ (initial population)}$$

$$a_n = 1.1 \times a_{n-1} - 80 \text{for} n \geq 1$$

Computing the fish population for the first 5 months:

$$a_0 = 1000$$

$$a_1 = 1.1 \times 1000 - 80 = 1020$$

$$a_2 = 1.1 \times 1020 - 80 = 1122 - 80 = 1042$$

$$a_3 = 1.1 \times 1042 - 80 = 1146.2 - 80 = 1066$$

$$a_4 = 1.1 \times 1066 - 80 = 1172.6 - 80 = 1092$$

$$a_5 = 1.1 \times 1092 - 80 = 1201.2 - 80 = 1121$$

Therefore, after 5 months, there will be 1121 fish in the lake.